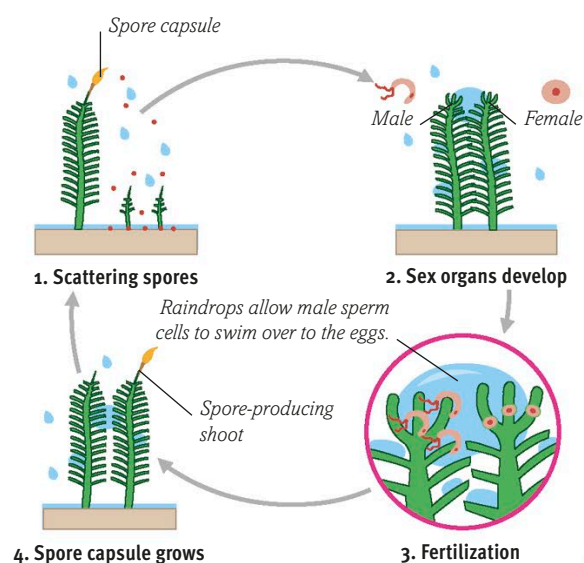


Nonflowering plants

The most ancient land plants on Earth are the nonflowering plants that evolved hundreds of millions of years ago. The earliest were simple plants such as liverworts, mosses, and hornworts, which grow in damp places to avoid drying out. Ferns are more complex but still have to live in moist environments. Instead of producing seeds, almost all nonflowering plants reproduce using tiny spores, which are carried away by wind or water. Only the gymnosperms, a group of nonflowering plants that includes conifers, produce seed-bearing cones instead.

Moss life cycle

A moss's spore capsule releases spores into the wind. When they land, they grow into leafy shoots with tiny sex organs. When it rains, male sperm cells are able to reach eggs and fertilize them. Each fertilized egg grows a new shoot, and the cycle continues.



Food factory ▶ The green leaflike part of this moss, called the gametophyte, produces food using energy from sunlight. It does not contain veins to transport water and nutrients, but its surface is so thin that these simply soak through.

